

Q1 $\frac{6}{13} \begin{pmatrix} -34 \\ -6 \end{pmatrix} + \frac{7}{13} \begin{pmatrix} 18 \\ 7 \end{pmatrix} = \begin{pmatrix} -6 \\ 1 \end{pmatrix}$ (C)

Q2

coeff $x^2 = -1 \times 4 = -4$

coeff $x = +1 \times -5 = -5$

constant $= -1 \times 6 = -6$

(A)

Q3

Total = $12!$

Unfavorable = $9! \times \binom{10}{3} \times 3!$

x _ x _ x _ x _ x _ x _ x _ x _ x _ x

choose positions

$\Rightarrow \text{Prob} = 1 - \frac{9! \times \binom{10}{3} \times 3!}{12!} = \frac{5}{11}$

arrange molecules

(B)

Q4

$X \sim B(10, 0.4)$

$P(X < 3 | X > 1) = \frac{P(X < 3 \cap X > 1)}{P(X > 1)}$

$= \frac{P(X = 2)}{1 - P(X \leq 1)}$

$= \frac{\binom{10}{2} (0.4)^2 (0.6)^8}{1 - 0.6^{10} - \binom{10}{1} (0.4) (0.6)^9} \approx 0.1268$

(B)

Q5

$$\int_x^{4x} \frac{1}{t + \sqrt{t}} dt$$

$$= \int_x^{4x} \frac{1}{\sqrt{t}(\sqrt{t} + 1)} dt$$

$$= \int_{\sqrt{x}}^{2\sqrt{x}} \frac{2}{u+1} du$$

$$= \left[2 \ln |u+1| \right]_{\sqrt{x}}^{2\sqrt{x}}$$

$$= 2 \ln \left(\frac{2\sqrt{x} + 1}{\sqrt{x} + 1} \right)$$

$$u = \sqrt{t}$$

$$2du = \frac{1}{\sqrt{t}} dt$$

t	x	4x
u	$\sqrt{x}$	$2\sqrt{x}$

(A)

Q6

Be careful of vector orientation!

$$\cos(180^\circ - \alpha) = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}| |\underline{b}|}$$

$$\alpha = 180^\circ - \arccos \left(\frac{17871}{187 \cdot 781} \right)$$
$$\approx 97^\circ$$

(A)

Q7

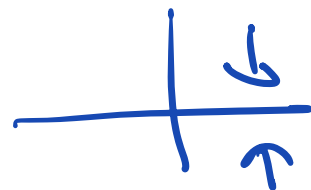
$$y = 3 \cos x + \pi$$



Reflection about y-axis:

$$\begin{aligned} y &= 3 \cos(-x) + \pi \\ &= 3(\pi - \cos x) + \pi \\ &= -3 \cos x + 4\pi \end{aligned}$$

Dilate vertically by $\frac{1}{2}$



$$y = -\frac{3}{2} \cos x + 2\pi$$

Translate down by $\frac{3\pi}{4}$

$$y = -\frac{3}{2} \cos x + \underbrace{(2\pi - \frac{3\pi}{4})}_{\frac{5\pi}{4}}$$

(D)

Q8

Try: (2,1)

A: $\frac{2^2 - 1^2}{4 - 1} = 1$ More

B: ~~$\frac{1^2 - 2^2}{4 - 1} = -1$~~

C: ~~$\frac{1^2 - 2^2}{4 + 1} = -\frac{3}{5}$~~

D: $\frac{2^2 - 1}{4 + 1} = \frac{3}{5}$ Less

(D)

Q9

$$\arcsin(\cos x)$$

$$= \arcsin\left(\sin\left(\frac{\pi}{2} - x\right)\right)$$

$$= \arcsin\left(-\sin\left(\frac{\pi}{2} - x + n\pi\right)\right) \text{ if } \frac{\pi}{2} \leq \frac{\pi}{2} - x \leq \frac{3\pi}{2}$$

$$\begin{cases} \sin(2\pi + x) = \sin x \\ \sin(\pi + x) = -\sin x \\ \text{For odd } n \\ \sin(n\pi + x) = -\sin x \end{cases}$$

$$= -\left(\frac{\pi}{2} - x + n\pi\right)$$

$$= x - n\pi - \frac{\pi}{2}$$

$$\arcsin(\sin x) = x$$

$$\text{if } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

$$\arcsin\left(\sin\left(\frac{\pi}{2} - x\right)\right) = x$$

$$\frac{\pi}{2} \leq \frac{\pi}{2} - x \leq \frac{3\pi}{2}$$

$$-\pi \leq -x \leq 0$$

$$0 \leq x \leq \pi$$

$$n\pi < x < (n+1)\pi$$

$$0 < x - n\pi < \pi$$

$$-\pi < n\pi - x < 0$$

$$-\frac{\pi}{2} < n\pi + \frac{\pi}{2} - x < \frac{3\pi}{2}$$

(D)

Q10

$$\text{Total: } \binom{100}{4}$$

Favorable:

$$d=1 \rightarrow (1, 3, 97, 100), (2, 3, 96, 100), \dots, (97, 98, 99, 100)$$

$$d=2 \rightarrow 94 = 100 - 3 \times 2$$

$$d=3 \rightarrow 91 = 100 - 3 \times 3$$

$$d=4 \rightarrow (1, 5, 9, 13), \dots, (88, 92, 96, 100)$$

⋮

$$d=33 \rightarrow (1, 34, 67, 100)$$

$$\text{Add: } 33 \times 100 - 3(1 + 2 + \dots + 33)$$

$$= 3300 - 3 \times \frac{33 \times 34}{2}$$

$$= 1617$$

$$\text{Final ans: } \frac{1617}{\binom{100}{4}} = \frac{1}{2425}$$

⑦

Question 11

$$\begin{aligned} \text{a) } \int \frac{8}{\sqrt{9-4x^2}} dx &= \int \frac{8}{2\sqrt{\frac{9}{4}-x^2}} dx \\ &= 4 \arcsin\left(\frac{2x}{3}\right) + C \end{aligned}$$

$$\text{b) } 2\sin\theta = 1 + \cos\theta$$

$$\cos\theta - 2\sin\theta = -1$$

$$\begin{aligned} \cos\theta - 2\sin\theta &\equiv R\cos(\theta + \alpha) \\ &= R\cos\theta\cos\alpha - R\sin\theta\sin\alpha \end{aligned}$$

$$R\cos\alpha = 1 \quad \text{--- ①}$$

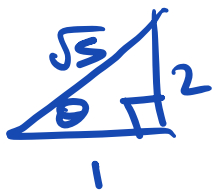
$$R\sin\alpha = 2 \quad \text{--- ②}$$

$$\begin{aligned} \text{①}^2 + \text{②}^2 : R^2(\cos^2\alpha + \sin^2\alpha) &= 1^2 + 2^2 \\ R &= \sqrt{5} \end{aligned}$$

$$\begin{aligned} \text{②} \div \text{①} : \tan\alpha &= 2 \\ \alpha &= \arctan 2 \end{aligned}$$

$$0 \leq \theta \leq 2\pi$$

$$\cos(\theta + \arctan 2) = \frac{1}{\sqrt{5}} \quad \arctan 2 \leq \theta + \arctan 2 \leq 4\pi + \dots$$

Note:  for triangle
 $\arctan 2 = \arcsin \frac{1}{\sqrt{5}}$

don't tell
 arnyuki7

$\frac{\textcircled{S}}{\textcircled{T}} \mid \frac{A}{C}$

$$\Theta + \arcsin \frac{1}{\sqrt{5}} = \pi - \arcsin \frac{1}{\sqrt{5}}, \pi + \arcsin \frac{1}{\sqrt{5}}$$

$$\Rightarrow \Theta = \pi - 2\arcsin \frac{1}{\sqrt{5}}, \pi.$$

c) $f(x) = \arctan x - x$

$$f'(x) = \frac{1}{1+x^2} - 1$$

$$= \frac{1 - 1 - x^2}{1+x^2}$$

$$= -\frac{x^2}{1+x^2}.$$

Since $x^2 \geq 0$ (and $1+x^2 > 0$),

$f'(x) \leq 0$, which means

$f(x)$ is monotonically decreasing on its entire domain, and is therefore one-to-one.

Hence $f(x)$ is invertible.

Note $f(x)$ attain only one stationary point (at $x=0$), hence it will still satisfy the horizontal line test.

$$d) X \sim N(1200, 200)$$

$P(\text{1 light bulb fails within 1000 hr})$

$$= P(X \leq 1000)$$

$$= P\left(Z \leq \frac{1000 - 1200}{200}\right)$$

$$= P(Z \leq -1)$$

$$= 1 - P(Z \leq 1)$$

$$= 1 - 0.8413$$

$$= 0.1587 \rightarrow p$$

$Y \sim B(64, 0.1587)$, want

$$P(Y \geq 3) = 1 - 0.8413^{64} - \binom{64}{1} 0.8413^{63} \cdot 0.1587 \\ - \binom{64}{2} 0.8413^{62} \cdot 0.1587^2$$

$$\approx 0.999984$$

$$e) \quad \frac{x^2 - 2}{x^2 - 3} \leq 2$$

$$x^2 - 3 \neq 0$$

$$x \neq \pm\sqrt{3}$$

$$\frac{\cancel{x^2 - 3}^1}{\cancel{x^2 - 3}} + \frac{1}{x^2 - 3} \leq 2$$

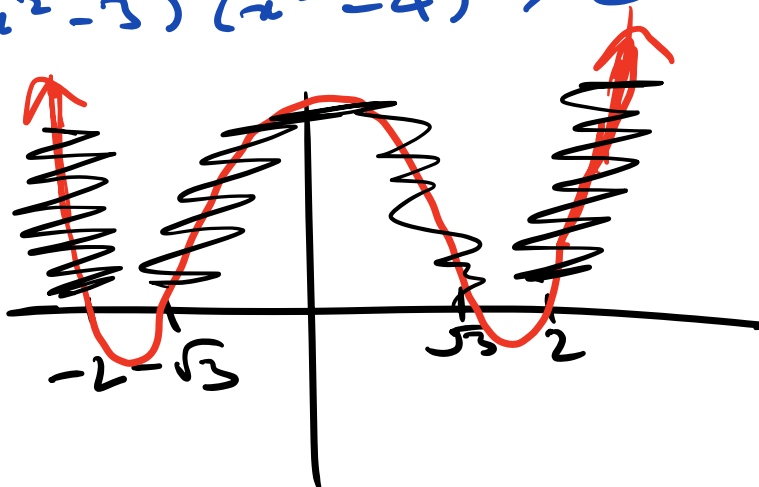
$$\frac{1}{x^2 - 3} \leq 1$$

$$x^2 - 3 \leq (x^2 - 3)^2$$

$$(x^2 - 3)^2 - (x^2 - 3) \geq 0$$

$$(x^2 - 3) [(x^2 - 3) - 1] \geq 0$$

$$(x^2 - 3)(x^2 - 4) \geq 0$$



$$\therefore x \leq -2, -\sqrt{3} < x < \sqrt{3}, x \geq 2$$

f) Consider

$$\begin{aligned} & (x+\pi)^3 - 2\pi(x+\pi)^2 + 5\pi^2(x+\pi) - \pi^3 \\ &= x^3 + (\dots)x^2 + (\dots)x + \pi^3 - 2\pi(\pi^2) + 5\pi^2(\pi) - \pi^3 \\ &= x^3 + \dots + 3\pi^3 \end{aligned}$$

Using above polynomial, we see

$$(x-\pi)(x-\pi)(x-\pi) = -\frac{3\pi^3}{1}.$$

Questão 12

a) $N = 2130 - (T + 440) [\ln(T + 440)]^2$

When $T = -40$, $\frac{dT}{dt} = -\frac{1}{2}$

$\Rightarrow$ When $T = 40$,

$$\frac{dN}{dt} = \frac{dN}{dT} \frac{dT}{dt}$$

$$= \frac{1}{2} \left([\ln(T + 440)]^2 + (440 + T) \frac{2 \ln(T + 440)}{T + 440} \right)$$

$$\approx 100.068$$

$\therefore$ Increasing at 100 (approx) tints per hour.

b)

$$\frac{dP}{dt} = k(P - 10)$$

$$k dt = \frac{dP}{P - 10}$$

$$\int k dt = \int \frac{dP}{P - 10}$$

$$kt = \ln |P - 10| + C$$

Since $P(1) = 1$, we require negative case for abs. value.

$$P(1) = 1$$

$$k = \ln 9 + C$$

— (1)

$$P(3) = 5$$

$$3k = \ln 5 + C$$

— (2)

$$\textcircled{1} - \textcircled{1}: \quad 2c = \ln \frac{9}{5}$$

$$c = \frac{1}{2} \ln \frac{9}{5}$$

sub into $\textcircled{1}$: $\frac{1}{2} \ln \frac{9}{5} = \ln 9 + c$

$$\therefore \frac{1}{2} \ln \frac{9}{5} = \ln |P-10| + \frac{1}{2} \ln \frac{9}{5} - \ln 9$$

For $P(6)$:

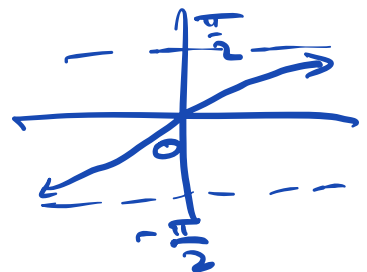
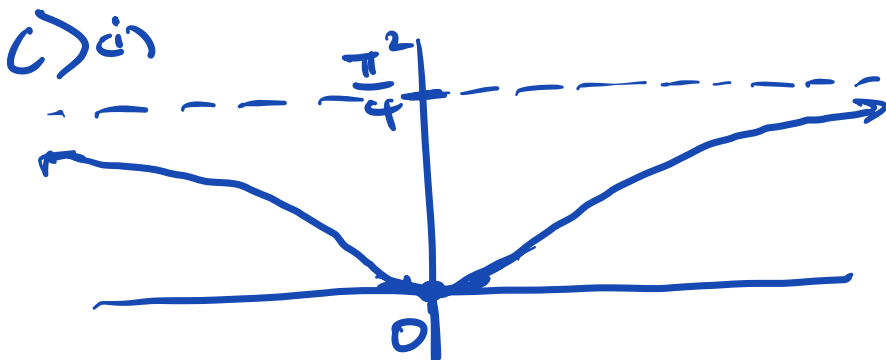
$$3 \ln \frac{9}{5} = \ln |P-10| + \frac{1}{2} \ln \frac{9}{5} - \ln 9$$

$$\frac{5}{2} \ln \frac{9}{5} - \ln 9 = \ln(10-P)$$

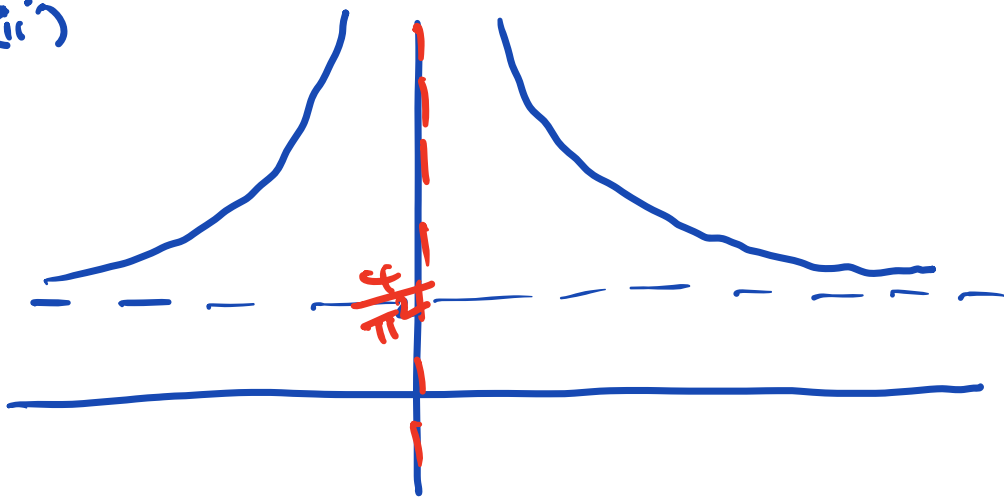
$$\frac{e^{\frac{5}{2} \ln \frac{9}{5}}}{e^{\ln 9}} = 10-P$$

$$P = 10 - \frac{\left(\frac{9}{5}\right)^{\frac{5}{2}}}{9}$$

$$\approx 9.517$$



(ii)



d) (i) Area of region I

$$= \int_0^1 \sqrt{x} - x^3 dx$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} - \frac{x^4}{4} \right]_0^1$$

$$= \frac{5}{12}$$

$$\text{Prob of landing in I} = \frac{\text{Area of I}}{\underbrace{\text{Area of unit square}}_{=1 \times 1 = 1}} = \frac{5}{12}$$

(ii) Following similar idea,

$$\frac{29}{75} = \int_0^1 \sqrt{x} - x^r dx$$

$$= \left[\frac{2x^{\frac{3}{2}}}{3} - \frac{x^{r+1}}{r+1} \right]_0^1$$

$$\frac{29}{75} = \frac{2}{3} - \frac{1}{r+1}$$

$$\frac{1}{r+1} = \frac{7}{25}$$

$$r = \frac{18}{7} \quad \text{one eight / seven}$$

$$e) \quad 0 = \left(\frac{a+2b}{4} \right) \cdot \left(\frac{2a-b}{3} \right) \quad (\text{expanded})$$

$$0 = 12 + 2a^2 + \overset{3ab}{\cancel{4ab}} - ab - 2b^2 \quad \text{--- ①}$$

$$\left| \left(\frac{a+2b}{4} \right) \right|^2 = \left| \left(\frac{2a-b}{3} \right) \right|^2$$

$$(a+2b)^2 + 16 = (2a-b)^2 + \overset{9}{\cancel{9}}$$

$$\overset{+7}{\cancel{+7}} = (2a-b)^2 - (a+2b)$$

$$= [(2a-b) - (a+2b)] [(2a-b) + (a+2b)]$$

$$= (a-3b)(3a+b)$$

$$= 3a^2 - 8ab - 3b^2 \quad \text{--- ②}$$

$$\textcircled{1} \times 3 \parallel -36 = 6a^2 + 9ab - 6b^2 \quad \text{--- ③}$$

$$\textcircled{2} \times 2 \parallel \overset{24}{\cancel{24}} = 6a^2 - 16ab - 6b^2 \quad \text{--- ④}$$

$$\textcircled{3} - \textcircled{4} - 50 = 25ab$$

$$ab = -2 \quad \text{---} \quad \textcircled{5}$$

$$\textcircled{1}: -12 = 2a^2 + 3ab - 2b^2 \quad \text{sub in } \textcircled{5}$$

$$= 2a^2 - 6 - 2b^2$$

$$-6 = 2a^2 - 2b^2$$

$$-3 = a^2 - b^2$$

$$-3 = a^2 - \frac{4}{a^2}$$

$$0 = a^4 + 3a^2 - 4$$

$$= (a^2 - 1)(a^2 + 4)$$

$$a = \pm 1$$

$$b = -\frac{2}{a}$$



no solutions

Question 13

a) i) m consecutive integers can always be written in the form $n+1, n+2, \dots, n+m$. for some integer $n \geq 0$

$$\begin{aligned} \text{But: } \binom{n+m}{m} &= \frac{(n+m)!}{m! n!} & n! &= n(n-1)! \\ & & &= n(n-1)(n-2)! \\ &= \frac{(n+m)(n+m-1) \dots (n+2)(n+1) \cancel{n!}}{\cancel{m!} \cancel{n!}} \\ &= \underbrace{(n+m)(n+m-1) \dots (n+2)(n+1)}_{m!} \end{aligned}$$

The numerator is the product of the aforementioned consecutive integers. But we know $\binom{n+m}{m}$ is always an integer.

Therefore, the required product of integers must be divisible by $m!$.

(ii) For $n=1$:

$$(3n)! = 3! = 6 = 6^1 \times 1. \quad \checkmark$$

Assume for $n=k$:

$$(3k)! = 6^k M \quad \text{for another } M \in \mathbb{Z}.$$

Then for $n=k+1$,

$$(3(k+1))!$$

$$= (3k+3)!$$

$$= (3k+3)(3k+2)(3k+1)(3k)!$$

$$= (3k+3)(3k+2)(3k+1)(6^k M) \quad (\text{assumed})$$

$$= (6N)(6^k M) \quad \text{for some } N \in \mathbb{Z}$$

because $3k+3$, $3k+2$, $3k+1$ are 3 consecutive integers, hence by part (i) is divisible by $3! = 6$

$$= 6^{k+1} (NM)$$

which is divisible by 6^{k+1} .

Hence the statement holds by mathematical induction.

b) (i)

$$\underline{r}_A(t) = \underline{r}_B(t_0)$$

$$U_{t_0} \cos \alpha = v(t_0 - T) \cos \beta$$

— ①

$$U_{t_0} \sin \alpha - \frac{gt_0^2}{2} = v(t_0 - T) \sin \beta - \frac{g(t_0 - T)^2}{2}$$

— ②

Rearrange ①:

$$u_{to} \cos \alpha = v_{to} \cos \beta - v_T \cos \beta$$

$$v_T \cos \beta = v_{to} \cos \beta - u_{to} \cos \alpha$$

$$= t_o (v \cos \beta - u \cos \alpha)$$

$$t_o = \frac{v_T \cos \beta}{v \cos \beta - u \cos \alpha} \quad \leftarrow \textcircled{3}$$

(ii) Range $\textcircled{2}$:

$$u_{to} \sin \alpha - \frac{gt_o^2}{2} = v_{to} \sin \beta - v_T \sin \beta$$

$$- \frac{gt_o^2}{2} + gt_o T - \frac{gT^2}{2}$$

$$v_T \sin \beta + \frac{gT^2}{2} = v_{to} \sin \beta - u_{to} \sin \alpha + gt_o T$$

$$= t_o (v \sin \beta - u \sin \alpha + gT)$$

$$t_o = \frac{v_T \sin \beta + \frac{gT^2}{2}}{v \sin \beta - u \sin \alpha + gT} \quad \leftarrow \textcircled{4}$$

$$\therefore \frac{\cancel{v_T} \cos \beta}{v \cos \beta - u \cos \alpha} = \frac{\cancel{v_T} \sin \beta + \frac{gT^2}{2}}{v \sin \beta - u \sin \alpha + gT}$$

can cancel T as $T > 0$
(C is cancelled after A)

$$V \cos \beta (V \sin \beta - U \sin \alpha + gT)$$

$$= (V \cos \beta - U \cos \alpha) \left(V \sin \beta + \frac{gT}{2} \right)$$

$$\cancel{V^2 \cos \beta \sin \beta} - UV \sin \alpha \cos \beta + gVT \cos \beta$$

$$= \cancel{V^2 \cos \beta \sin \beta} - UV \cos \alpha \sin \beta$$

$$+ \frac{gVT \cos \beta}{2} - \frac{gUT \cos \alpha}{2}$$

$$\frac{gVT \cos \beta + gUT \cos \alpha}{2} = UV \sin \alpha \cos \beta - UV \cos \alpha \sin \beta$$

$$gT(V \cos \beta + U \cos \alpha) = 2UV (\sin \alpha \cos \beta - \cos \alpha \sin \beta)$$

$$T = \frac{2UV \sin(\alpha - \beta)}{U \cos \alpha + V \cos \beta}$$

c) (i) Cases:

- Right next to each other: 2×19

- Gap 1: 2×18

- Gap 2: 2×17



$\Rightarrow 108$ ways

(ii) $P(\leq 2 \text{ spaces})$

$$= P(\leq 2 \text{ spaces} | \text{same row}) P(\text{same row})$$

~~$P(\leq 2 \text{ spaces}) \text{ different cars}) \neq (\text{different cars})$~~

$w_{rel}(i) = \frac{108}{\binom{20}{2} \times 2!} \times \frac{19}{59} \rightarrow {}^{20}P_2$

$$= \frac{27}{295}$$

Replacement

(iii) $P(\text{car in a XO spot } 1 \leq 2 \text{ spaces apart})$
 $\text{car in } 10, 20, 30 + \text{car in } 20, 40, 60$

$$\frac{P(\text{car in } 10, 20, 30, 40, 50, \text{ or } 60 \cap \leq 2 \text{ years apart})}{P(\leq 2 \text{ years apart})} \leftarrow \text{we (ii)}$$

Favorable action:

— Case 1: car in 20, 40, 60.

If car is 20, then other car not be in

19, 18, or 17.

→ 3×2

Cheris fur
las porties

Arrange two
cars

Cars at 40 & 60 are identical

→ Total is $3 \times 3 \times 2 = 18$.

- Case 2: car in 10, 30, 50.

This is the same idea, except the other car can now be on the other side. e.g.

7 8 9 10 11 12 13

→ Total is $2 \times 18 = 36$

There combined favorable outcomes = 54.

$$\therefore \text{Prob} = \frac{\frac{54}{2! \binom{60}{2}}}{\frac{27}{295}} = \frac{1}{6}$$

ALTERNATE FASTER APPROACH

Numerator: # favorable outcomes (i.e. 10, 20, ..., & 60 miles)

allowing 2 spaces apart

$$= 54$$

Denominator: # outcomes where the cars are 2 spaces apart.

$$= 3 \times 108 = 324 \rightarrow \frac{54}{324} = \frac{1}{6}$$

Question 14

a) Let n = number of tickets sold.

$$\text{Attendance} = n + X,$$

$$X \sim B(n, 0.5)$$

(X is the number of people who bring a friend).

$$\text{We want } P(n + X > 60) < 0.01$$

$$P(X > 60 - n) < 0.01$$

$$1 - P(X \leq 60 - n) < 0.01$$

$$P(X \leq 60 - n) > 0.99$$

Approximate X by $N(\frac{n}{2}, \frac{n}{4})$, $n=n$
 $p=\frac{1}{2}$

then if $Z \sim N(0,1)$, we have

$$P\left(Z \leq \frac{(60-n) - \frac{n}{2}}{\sqrt{\frac{n}{4}}}\right) > \text{0.99}$$

table

$$\frac{(60-n) - \frac{n}{2}}{\sqrt{\frac{n}{4}}} > 2.33$$

$$60 - \frac{3n}{2} > 2.33 \sqrt{\frac{n}{4}}$$

$$\frac{3n}{2} + 2.33\sqrt{\frac{n}{4}} - 60 < 0$$

$$1.5n + 1.165\sqrt{n} - 60 < 0$$

For equality,

$$\sqrt{n} = \frac{-1.165 \pm \sqrt{1.165^2 + 360}}{3}$$

reject negative case as $\sqrt{n} > 0$

$$n = (5.94813\dots)^2$$

$$= 35.38028\dots$$

$\Rightarrow$ At least 36 people.

$$b) (x^2 + 8x + 7)^{11}$$

$$= [(x^2 + 2x + 1) + \underline{6(x+1)}]^{11}$$

$$= \binom{11}{0} (x^2 + 2x + 1)^{11} + \binom{11}{1} (x^2 + 2x + 1)^{10} [6(x+1)]$$

$$+ \binom{11}{2} (x^2 + 2x + 1)^9 [6(x+1)]^2$$

+ ...

We want to consider the coefficient of x^3 , but

only the remainder when divided by 6.

TRICK: Irrespective of the coefficient of x^3 in $\binom{11}{1}(x^2+2x+1)^{10}[6(x+1)]$,

$$\binom{11}{2}(x^2+2x+1)^9[6(x+1)]^2, \dots,$$

its remainder will always be 0,

because 6 is a factor of the entire term.

Hence we only need the coeff of x^3 in the first term $(x^2+2x+1)^{11} = (x+1)^{22}$

$$\begin{aligned}\Rightarrow \binom{22}{3} &= \frac{22!}{3!19!} \\ &= \frac{22 \times 21 \times 20}{6} \\ &= 22 \times 7 \times 10 \\ &= 1540\end{aligned}$$

$$\text{Know: } 1500 = \frac{3000}{2} = 6 \times 250$$

$\therefore$ The remainder is the same as the remainder for $40 \div 6$, which is 4.

c)

$$p + q + r = 0$$

$$\hat{p} + \hat{q} + \hat{r} = 0$$

(i)

$$r = -(p + q)$$

$$\hat{r} = \frac{r}{1-r}$$

$$= \frac{-(p+q)}{1-(p+q)}$$

$$= -\frac{1}{1-p-q} (p+q)$$

$$\Rightarrow \lambda = -\frac{1}{1-p-q}$$

$$(ii) \quad 0 = \hat{p} + \hat{q} + \hat{r}$$

$$= p + q - \frac{p+q}{1-p-q}$$

$$= \frac{p}{1-p} + \frac{q}{1-q} - \frac{p}{1-p-q} - \frac{q}{1-p-q}$$

$$p \left(\frac{1}{1-p-q} - \frac{1}{1-p} \right) = q \left(\frac{1}{1-q} - \frac{1}{1-p-q} \right)$$

$$p \left(1 - \frac{1-p-q}{1-p} \right) = q \left(\frac{1-p-q}{1-q} - 1 \right)$$

$$\hat{P} \left(\frac{|P| - |R+Q|}{|P|} \right) = \hat{Q} \left(\frac{|R+Q| - |Q|}{|Q|} \right).$$

(iii) Case 1 $\hat{P}, \hat{Q}$ parallel:

Suppose $\hat{P}, \hat{Q}$ parallel, then they are scalar multiples of one another, BUT they are both unit vectors.

So:

Case $\hat{P} = \hat{Q}$ $\hat{r} = -\hat{P} - \hat{P}$ $= -2\hat{P}$	$\hat{P} = -\hat{Q}$ $\hat{r} = -\hat{P} + \hat{P}$ $= 0$
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Both cases are contradictions because unit vectors can only have magnitude 1, and this suggests magnitude 2 or 0.

Case 2: $\hat{P}, \hat{Q}$ non-parallel

From (ii),

$$(|P| - |R+Q|) \hat{P} = (|R+Q| - |Q|) \hat{Q}.$$

Because $\hat{P}$ and $\hat{Q}$ are not parallel, they are not linearly independent, hence can't be non-zero scalar multiples of each other.

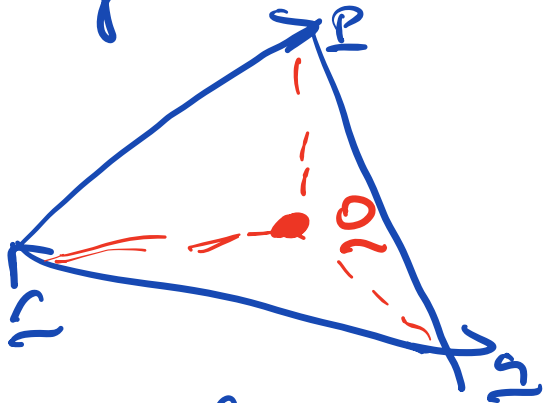
$$\therefore |\vec{P}| - |\vec{P} + \vec{Q}| = 0$$

$$\Rightarrow |\vec{P}| = |\vec{P} + \vec{Q}| = |\vec{r}|$$

AND $|\vec{P} + \vec{Q}| - |\vec{Q}| = 0$

$$\Rightarrow |\vec{Q}| = |\vec{P} + \vec{Q}| = |\vec{r}|$$

$\Rightarrow$ hence all sides are equal, but also the sum of vectors $\vec{P} + \vec{Q} + \vec{r} = \vec{0}$.



$\Rightarrow \vec{P}, \vec{Q}, \vec{r}$ form sides of an equilateral triangle.

d) i)

$P(k) = 0$ otherwise is obvious because can't have negative # of coins left and can't have more coins than what you started with.

(Also can't finish on 0 coins.)

For $k=1, 2, \dots, n$, first note it does not matter which pocket has remaining coins.

(Because: if he ends up with k coins in his left pocket, by simply inverting which pocket he draws from, he will have the same amount of coins in the right pocket.)

$\therefore$ Assume which coins end up in left pocket

$P(k \text{ in left pocket}) = P((n-1) \text{ coins taken out of right pocket AND } n-k \text{ coin taken out of left pocket IN ANY ORDER}) \times P(\text{out of right pocket})$

$$= \binom{(n-1) + (n-k)}{n-k} \left(\frac{1}{2}\right)^{n-k} \left(\frac{1}{2}\right)^{n-1} \times \frac{1}{2}$$

$$= \binom{2n-k-1}{n-k} \frac{1}{2^{2n-k-1}} \times \frac{1}{2}$$

$$P(k \text{ is right packet}) = P(k \text{ is left packet})$$

$$\Rightarrow P(k) = \binom{2n-k-1}{n-k} \frac{1}{2^{2n-k-1}}$$

(ii)

$$2(n-k) P(k) = (n-k) \binom{2n-k-1}{n-k} \frac{1}{2^{2n-k-2}}$$

$$= \cancel{(n-k)} \frac{(2n-k-1)!}{\cancel{(n-k)!} (n-1)!} \frac{1}{2^{2n-k-2}}$$

$$= \frac{(2n-k-1)!}{(n-k-1)! (n-1)!} \frac{1}{2^{2n-k-2}}$$

$$= (2n-k-1) \frac{(2n-(k+1)-1)!}{(n-k-1)! (n-1)!} \frac{1}{2^{2n-k-1}}$$

$\begin{aligned} &= 2n - (k+1) - 1 \\ &= 2n - k - 2 \end{aligned}$

$$= (2n-k-1) \binom{2n-(k+1)-1}{n-(k+1)} \frac{1}{2^{2n-(k+1)-1}}$$

$\begin{aligned} &= n - (k+1) \end{aligned}$

$$\sum_k c a_n = c \sum_k a_n = (2n-k-1) P(k+1) \quad \begin{aligned} &\sum (a_k + b_k) \\ &= \sum_k a_k + \sum_k b_k \end{aligned}$$

$$(iii) \quad \sum_{k=1}^n 2(n-k) P(k) = \sum_{k=1}^n (2n-k-1) P(k+1)$$

$$2n \sum_{k=1}^n P(k) - 2 \sum_{k=1}^n k P(k) = 2n \sum_{k=1}^n P(k+1) - \sum_{k=1}^n (k+1) P(k+1)$$

BUT: $\sum_{k=1}^n P(k) = 1$

AND: $\sum_{k=1}^n k P(k) = E(X)$ by definition.

$\therefore LHS = 2n - 2E(X)$

Then: $\sum_{k=1}^n P(k+1)$

$$= P(2) + P(3) + \dots + P(n) + \overbrace{P(n+1)}^{=0}$$

$$= -P(1) + [P(1) + P(2) + P(3) + \dots + P(n)]$$

$$= 1 - P(1)$$

Similarly: $\sum_{k=1}^n (k+1) P(k+1)$

$$= 2P(2) + 3P(3) + \dots + nP(n) + \overbrace{(n+1)P(n+1)}^{=0}$$

$$= -1P(1) + [1P(1) + 2P(2) + 3P(3) + \dots + nP(n)]$$

$$= E(X) - P(1)$$

$$\Rightarrow RHS = 2n(1 - P(1)) - (E(X) - P(1))$$

$$= 2n - 2nP(1) - E(X) + P(1)$$

Putting everything together,

$$\begin{aligned}
 2n - 2E(X) &= 2n - 2n P(1) - E(X) + P(1) \\
 -E(X) &= (-2n + 1) P(1) \\
 E(X) &= (2n - 1) P(1).
 \end{aligned}$$

$$(iv) \quad P(1) = \binom{2n-2}{n-1} \frac{1}{2^{2n-2}}$$

$$\Rightarrow E(X) = (2n-1) \frac{(2n-2)!}{[(n-1)!]^2} \frac{1}{2^{2n-2}}$$

$$= \frac{(2n-1)!}{[(n-1)!]^2} \frac{1}{2^{2n-2}}$$

$$= \frac{(2n)!}{(n!)^2} \frac{n^2}{2n} \frac{1}{2^{2n-2}}$$

$$= \binom{2n}{n} \frac{n}{2^{2n-1}}$$

We know that $E(X) \leq n$ always holds, because n is the max value of this random variable.

$$\therefore \binom{2n}{n} \frac{n}{2^{2n-1}} \leq n \Rightarrow \binom{2n}{n} \leq 2^{2n-1}.$$